



User Stories and Acceptance Criteria

TASCO Insurance × VETC Growth Platform — Business & Product

iorta TechNXT

Version 1.0 · 07 October 2026

Contents

0. Conventions

EP-01 Data foundation and repair

EP-02 Lead prioritisation

EP-03 Renewal journeys (TASCO book)

EP-04 New-business journeys and moments of truth

EP-05 AI voice bot

EP-06 Telesales closing

EP-07 Customer app: renew and self-service

EP-08 Value beyond discount and cross-sell

EP-09 Partner channel

EP-10 Fleet / B2B

EP-11 Claims FNOL

EP-12 Privacy and consent

EP-13 Rule governance

EP-14 Identity, access and audit

EP-15 Operations, insights and UX

Document	Version	Date	Owner	Status
User Stories and Acceptance Criteria	1.0	07 October 2026	iorta TechNXT — Business Analysis	Issued for submission

TASCO Insurance × VETC Motor Insurance Growth Platform · prepared by iorta TechNXT

0. Conventions

- **IDs.** Stories are `US-nnn`, epics are `EP-nn`, personas `PC-n` (customer) and `PS-nn` (staff), as defined in doc 06. FR IDs refer to doc 02. References such as "doc 05" point to the companion documents listed in the Business and Product Documentation index (00-index.pdf).
- **Format.** As a ... I want ... so that ..., followed by acceptance criteria in Gherkin (Given / When / Then). Thresholds quoted in criteria are the **current rule values** from `config/rules/*.json`. If the business changes a rule through maker-checker, the criteria follow the active rule, not the literal number.
- **Priority** uses MoSCoW, as in doc 02.
- **Gherkin background used throughout:** "today" is the platform clock (`SIM_TODAY` in UAT), the timezone is ICT (UTC+7), and the rule sets in `config/rules` are active at version 1.
- **Automated acceptance tests.** Each scenario is one automated test named "US-*nnn* · scenario title" in `test/functional/*.test.js` (124 scenarios: 121 automated, 3 verified manually or in UAT: US-055 fleet view, US-076 language toggle, US-077 physical QR scan). All automated scenarios pass. Results are recorded in docs/quality/TASCO-Test-Cases-and-Results.xlsx. Scenario titles marked "(target behaviour)" keep their original names for traceability; the behaviour is now implemented.

0.1 Epic overview

Epic	Name	Track	Stories
EP-01	Data foundation and repair	T2	US-001 – US-006
EP-02	Lead prioritisation	T2	US-007 – US-011
EP-03	Renewal journeys (TASCO book)	T3	US-012 – US-016
EP-04	New-business journeys and moments of truth	NB	US-017 – US-022
EP-05	AI voice bot	T1	US-023 – US-031
EP-06	Telesales closing	T1	US-032 – US-035
EP-07	Customer app: renew and self-service	T3	US-036 – US-042
EP-08	Value beyond discount and cross-sell	T4	US-043 – US-048
EP-09	Partner channel	NB	US-049 – US-053
EP-10	Fleet / B2B	NB	US-054 – US-055
EP-11	Claims FNOL	T4	US-056 – US-058
EP-12	Privacy and consent	PL	US-059 – US-062
EP-13	Rule governance	PL	US-063 – US-068
EP-14	Identity, access and audit	PL	US-069 – US-072
EP-15	Operations, insights and UX	PL	US-073 – US-077

EP-01 Data foundation and repair

US-001: Ingest a partner list into golden profiles

As a data steward (PS-06) I want to upload a partner or telesales list of vehicles so that duplicates merge into one profile per plate and bad records are quarantined. **FR:** FR-001, FR-002, FR-003, FR-004, FR-011 ·

Priority: M

```
Scenario: Duplicate records for the same vehicle merge into one golden profile
  Given a VETC account record with plate "30a-123.45" and phone "+84912345678"
  And a partner_bank record with plate "30A 12345" and phone "0912 345 678"
  When I POST both records to /api/data/ingest with source "partner_bank"
  Then exactly one profile with id "30A12345" and plate "30A-123.45" exists
  And its phone is "0912345678"
  And its sources include "vetc_account" and "partner_bank"
```

```
Scenario: Invalid plate is rejected and raised as a DQ issue
  Given a record with plate "30-12345"
  When the batch is ingested
  Then the response reports rejected = 1
  And a DQ issue of type "invalid_plate" is open for that record id
```

```
Scenario: Batch size limit
  When I post a batch of 5,001 records
  Then the request is rejected with a validation error
```

US-002: See how reliable an expiry date is

As a campaign manager (PS-03) I want each vehicle's expiry date to show its confidence and evidence so that I don't run sales journeys on guesses. **FR:** FR-005, FR-008 · **Priority:** M

```
Scenario: Verified certificate wins
  Given a vehicle has a verified certificate expiring 2027-03-01 and an inspection-cycle estimate of 2027-03-10
  When the profile is built
  Then policy.expiryDate is "2027-03-01"
  And expiryMethod is "verified_certificate" with confidence 1.0
```

```
Scenario: Agreeing weak evidence is corroborated
  Given only an inspection-cycle estimate (0.5) and a tag-anniversary estimate (0.25) within 21 days of each other
  When the profile is built
  Then the chosen method is "inspection_cycle"
  And confidence is 0.65
```

```
Scenario: Lineage shows the evidence
  When I GET /api/customers/{id}/lineage
  Then I see a field entry "policy.expiryDate" with its source and confidence
```

US-003: Correct an expiry with evidence

As a data steward (PS-06) I want to correct a vehicle's expiry and insurer when a customer sends proof so that reminders go out at the right time. **FR:** FR-009, FR-097 · **Priority:** M

```

Scenario: Correction requires evidence and is audited
  Given I hold permission "profile:update"
  When I PATCH /api/customers/30A12345/expiry with expiryDate "2027-05-10", insurer "PVI" and
  evidence "photo of certificate via hotline"
  Then the profile expiry is "2027-05-10" with confidence 0.9
  And the audit trail contains "profile.expiry_corrected" with my user id and the evidence text
  And the lead is recomputed immediately

Scenario: Missing evidence
  When I omit "evidence"
  Then the request fails validation

```

US-004: Work the data-quality queue

As a data steward (PS-06) I want a queue of data-quality issues by type **so that** I can fix the records that block revenue first. **FR:** FR-007 · **Priority:** M

```

Scenario: Filter and resolve
  Given open DQ issues of types "reliable_expiry" and "conflicting_phone"
  When I GET /api/dq/issues?type=conflicting_phone
  Then only "conflicting_phone" issues are returned with a byType summary
  When I POST /api/dq/issues/{id}/resolve with resolution "confirmed primary number by call"
  Then the issue status is "resolved" with my id and timestamp

```

US-005: Platform-captured facts survive source reloads

As a campaign manager (PS-03) I want customer confirmations and bot findings to persist when VETC reloads its source data **so that** we never forget what a customer told us. **FR:** FR-010 · **Priority:** M

```

Scenario: Customer declaration beats a weaker reload
  Given a customer declared expiry "2027-01-15" (customer_declared, 0.75)
  When the nightly VETC batch re-ingests that plate with only a tag-anniversary estimate (0.25)
  Then the profile keeps expiry "2027-01-15" with method "customer_declared"

Scenario: Consent withdrawal survives reload
  Given the customer withdrew call consent in the consent centre
  When the source batch says callConsent = true
  Then the profile consent.call remains false

```

US-006: Infer vehicle category and premium

As a campaign manager (PS-03) I want the platform to infer the TNDS category when seats are missing **so that** quotes and premiums are right. **FR:** FR-006, FR-053 · **Priority:** M

```

Scenario: Declared seats beat toll class
  Given a vehicle with seatsDeclared 7 and tollClass 1, used personally
  Then its category is "car_6_11" with confidence 0.85
  And its lead premium is 873,400 VND incl. VAT for a 365-day term

Scenario: Toll class only
  Given a vehicle with tollClass 1 and no seats
  Then its category is "car_under6" with confidence 0.55 and a DQ issue "vehicle_category" is open

```

EP-02 Lead prioritisation

US-007: Call the right customer first

As a telesales agent (PS-01) I want a lead queue sorted by score with plain-language reasons **so that** I spend my time on the customers most likely to renew. **FR:** FR-012, FR-014, FR-017 · **Priority:** M

Scenario: Score, tier and reasons

Given a TASCO customer expiring in 10 days, expiry confidence 1.0, 8 app sessions, push enabled and call consent

When leads are computed

Then the score is between 0 and 100 and the tier is "hot" when score $\geq$ 70

And reasons list each factor with points, max and a "why" such as "expires in 10 days"

Scenario: Do-not-contact zeroes the score

Given the same customer is on DNC

Then the score is 0 and the NBA is "suppress"

Scenario: Filtering

When I GET /api/leads?tier=hot&maxDays=30&sort=score

Then only hot leads with daysToExpiry $\leq$ 30 are returned in descending score order

US-008: Know the next best action

As a campaign manager (PS-03) I want one recommended action per vehicle **so that** every contact is consistent and explainable. **FR:** FR-015 · **Priority:** M

Scenario Outline: First matching NBA rule wins

Given a lead with <facts>

Then nextBestAction.action is "<action>"

Examples:

facts	action
consent.dnc = true	suppress
ownerType = company	route_b2b
expiryConfidence = 0.3	verify_expiry
journey = new_vehicle, confidence 0.8	welcome_new_vehicle
days = 60, confidence 0.8	nurture
journey = lapsed_uninsured	urgent_recovery
tier hot, consent.call, hasPhone, days 20	voice_bot
tier warm, channels.app_push, days 20	digital_reminder
phone only, days 20	sms_reminder
no phone, no app, no Zalo, days 20	enrich

US-009: See only my region

As a regional telesales agent (PS-01, region "Hà Nội") I want the queue limited to my region **so that** I only work customers I am allowed to see. **FR:** FR-017, FR-093 · **Priority:** M

Scenario: Region scoping

Given I am "agent.hn" with region "Hà Nội" and no dashboard:read permission

When I GET /api/leads?region=TP. Hồ Chí Minh

Then only leads with region "Hà Nội" are returned

Scenario: Profile outside region

When I GET /api/customers/{id} for a profile in "TP. Hồ Chí Minh"

Then the response is 403 with policy "regional_data"

US-010: Stop selling to insured customers

As a customer (PC-1) I want reminders to stop as soon as I have renewed with TASCO **so that** I am not spammed. **FR:** FR-019, FR-030 · **Priority:** M

```
Scenario: Purchase stops journeys
  Given I have scheduled renewal touchpoints
  When a TNDs policy is issued to me
  Then my lead has no journey and NBA "insured"
  And any scheduled touchpoint executed later is "cancelled" with reason "already insured with TASCO"
```

US-011: Re-score after a rule change

As a campaign manager (PS-03) I want to re-score all leads on demand **so that** an approved scoring change takes effect immediately. **FR:** FR-018 · **Priority:** S

```
Scenario: Recompute
  Given I hold "leads:recompute"
  When I POST /api/leads/recompute
  Then every non-anonymised profile is re-evaluated with the active rules
  And the audit trail has "leads.recomputed" with the count
  And leads whose journey or tier changed have their touchpoints re-planned
```

EP-03 Renewal journeys (TASCO book)

US-012: Timely renewal reminders

As a TASCO customer (PC-1) I want reminders before my TNDs ends, through the VETC app first **so that** I renew on time without effort. **FR:** FR-020, FR-021, FR-022, FR-025, FR-029 · **Priority:** M

```
Scenario: Renewal cadence is planned from expiry
  Given my insurer is TASCO and my expiry is 60 days from today
  When my lead is computed
  Then touchpoints are planned at -45 (verify_expiry), -30 (first_reminder), -21 (value_reminder), -7 (urgent_reminder) and 0 (expiry_day)
  And a voice_bot step at -14 exists only if my tier is hot or warm
  And a telesales step at -3 exists only if my tier is hot

Scenario: App push first, then fallback
  Given a due first_reminder with channels app_push, zalo_zns, sms and I have push disabled but Zalo linked
  When the journey run executes
  Then the message is sent on zalo_zns, rendered in Vietnamese with my plate, expiry dd/mm/yyyy and a signed link
```

US-013: Respect contact hours and caps

As a compliance officer (PS-05) I want marketing contacts limited by time and frequency **so that** we comply with anti-spam rules and keep customer trust. **FR:** FR-023 · **Priority:** M

```
Scenario: Outside the contact window
  Given a due marketing touchpoint
  When the journey run executes at 21:15 ICT
  Then the touchpoint is not sent, with reason "outside allowed contact hours"
  And it stays scheduled (deferred) for the next run inside 08:00-20:00 ICT
```

```
Scenario: Weekly cap
  Given the customer received 3 marketing messages in the last 7 days
  When another marketing touchpoint is due
  Then it is skipped with reason "weekly contact cap reached"
```

```
Scenario: No marketing consent
  Given consent.marketing = false
  Then marketing messages are skipped with reason "no marketing consent"
```

US-014: Never send discount language

As a compliance officer (PS-05) **I want** any customer message containing discount wording to be blocked **so that** we never breach the ban on premium discounts. **FR:** FR-024 · **Priority:** M

```
Scenario: Send-time block
  Given a template rendered to text containing "giảm giá"
  When the journey engine sends it
  Then the message status is "blocked" with reason starting "copy guard:"
  And no provider call is made
```

```
Scenario: Diacritic-insensitive
  Given a text containing "giam gia", "giảm giá" or "cash back"
  Then it is also blocked
```

US-015: Service messages still reach the customer

As a customer (PC-1) **I want** the expiry-day notice even if I recently got marketing messages **so that** I know I am about to be uninsured. **FR:** FR-023, FR-025 · **Priority:** M

```
Scenario: Service message bypasses caps but not DNC
  Given I received 3 marketing messages this week
  When the expiry_day step (marketing = false) is due
  Then it is sent
  But if I am on DNC it is skipped with reason "customer is on do-not-contact list"
```

US-016: Escalate unresponsive hot customers to a human

As a telesales supervisor (PS-02) **I want** hot customers who ignore digital reminders to land in the telesales queue 3 days before expiry **so that** we don't lose the renewal. **FR:** FR-044 · **Priority:** M

```
Scenario: Journey escalation
  Given a hot TASCO-renewal lead with call consent and no purchase
  When the -3 day telesales step runs
  Then a handoff with outcome "journey_escalation" is created with talking points
  And a message log entry "[telesales task H0-...]" is recorded
```

EP-04 New-business journeys and moments of truth

US-017: Recover uninsured vehicles

As a lapsed owner (PC-4) **I want** to be told clearly that my car may be uninsured, with an easy way to fix it **so that** I avoid fines and failed inspection. **FR:** FR-013, FR-020, FR-022 · **Priority:** M

Scenario: Lapsed vehicle enters the recovery journey
 Given my expiry was 20 days ago with confidence 0.75
 When my lead is computed
 Then my journey is "lapsed_uninsured" and the NBA is "urgent_recovery"
 And a lapsed_notice (service) is planned for today on app_push, zalo_zns, sms
 And a voice_bot step is planned at +2 days if I am hot or warm, and telesales at +5 days if hot

Scenario: Lapsed too long is not assumed uninsured
 Given my expiry was 90 days ago
 Then I am not in "lapsed_uninsured" and the urgency reason says "likely renewed elsewhere"

US-018: Welcome new cars

As a new car buyer (PC-3) **I want** VETC to help me store my insurance and remind me **so that** I never lapse. **FR:** FR-013, FR-026 · **Priority:** M

Scenario: Tag activation triggers onboarding
 Given VETC emits "vetc.tag_activated" for my plate
 When POST /api/ecosystem/events is processed
 Then my lead is re-evaluated into journey "new_vehicle" (unless I am already a TASCO customer)
 And a welcome message (service) is planned at tag date +1, verify_expiry at +7 and value_reminder (marketing) at +14

US-019: Win customers insured elsewhere

As a private car owner insured elsewhere (PC-1) **I want** a reason to renew via VETC that is not price **so that** switching is worth it. **FR:** FR-016, FR-020, FR-025 · **Priority:** M

Scenario: Conquest copy is benefit-led and lawful
 Given my insurer is "PVI" and I expire in 30 days
 When first_reminder is due
 Then the "conquest_reminder" template is used with my top approved benefit (e.g. "cứu hộ giao thông 24/7")
 And the text contains no copy_guard phrase
 And a welcome message (service) is planned at tag date +1, verify_expiry at +7 and value_reminder (marketing) at +14

US-020: Inspection is a moment of truth

As a car owner (PC-1) booking an inspection (*đăng kiểm*) **I want** a check that my TNDS is valid **so that** I don't fail on the day. **FR:** FR-026, FR-062 · **Priority:** S

Scenario: Inspection booked with expiry soon
 Given my TNDS expires in 40 days
 When "vetc.inspection_booked" is received
 Then an "inspection_tnds_check" service message is sent via app_push or zalo_zns

Scenario: Expiry far away
 Given my TNDS expires in 200 days
 Then the trigger result is "condition not met" and nothing is sent

US-021: Remind when I am in the app with money

As a customer (PC-2) topping up my VETC wallet **I want** a renewal prompt at that moment **so that** I can pay in one tap. **FR:** FR-026 · **Priority:** S

```
Scenario: Wallet top-up within ±30 days of expiry
  Given my expiry is in 12 days and I have marketing consent
  When "vetc.wallet_topped_up" is received between 08:00 and 20:00 ICT
  Then a first_reminder push is sent
  And if I already received 1 marketing contact today the result is "no permitted channel"
```

US-022: Rehearse moments of truth

As a campaign manager (PS-03) **I want** to simulate VETC events from the console **so that** I can test triggers before the live event feed is connected. **FR:** FR-026, FR-108 · **Priority:** S

```
Scenario: Simulator
  Given I hold "journeys:run"
  When I choose a customer and event type "vetc.long_trip_started" in the Journeys screen
  Then the console shows matched triggers and per-trigger results
  And the audit trail has "ecosystem.event_handled"
```

EP-05 AI voice bot

US-023: Know I am talking to a bot from VETC

As a customer (PC-1) **I want** the caller to say upfront that it is VETC's automated assistant and will never ask for OTPs or payment **so that** I can trust the call. **FR:** FR-031 · **Priority:** M

```
Scenario: Opening line
  When a bot session starts
  Then the first bot line (key "intro") states it is VETC's automated assistant, the call is recorded, VETC never asks for OTP or payment by phone
  And it asks me to say my licence plate
  And it does not say my plate, name or expiry
```

US-024: Prove it's really my car before discussing it

As a customer (PC-1) **I want** to be asked for my plate first **so that** my data is not disclosed to the wrong person. **FR:** FR-032, FR-033 · **Priority:** M

```
Scenario: Spoken plate matches
  Given my plate is "30A-123.45"
  When I say "ba không A một hai ba bốn năm"
  Then the session is verified and the bot states my expiry using the masked plate "30A-***.45"

Scenario: Plate mismatch
  When I say "29A 999 99"
  Then the bot ends politely (line "plateMismatch") with outcome "plate_mismatch"
  And a DQ issue "plate_mismatch" is opened for the profile

Scenario: Plate not understood three times
  When I say three utterances with no recognisable plate
  Then the call ends with outcome "unverified"
```

US-025: Honest answer about price

As a price-sensitive customer (PC-1) I want a straight answer about price so that I don't feel tricked. FR: FR-035, FR-056 · Priority: M

Scenario: Price asked
 Given I am verified and my premium is 480,700 VND incl. VAT
 When I say "phí bao nhiêu tiền"
 Then the bot says TNSD premiums are set by regulation and identical at every insurer, states "480.700 đồng" incl. VAT
 And mentions one-tap renewal, instant e-certificate and 24/7 roadside assistance
 And the handoff (if any) flags priceAsked = true

US-026: Handle "is this a scam?"

As a sceptical customer (PC-4) I want the bot to explain how I can verify it so that I don't hang up on a genuine service. FR: FR-034 · Priority: M

Scenario: Scam concern at any point
 When I say "sao biết số tôi, lừa đảo à"
 Then the bot replies with the "trust" line and offers an in-app notice
 And signals.trustConcern = true is passed to telesales

US-027: Tell the bot I already renewed

As a customer insured elsewhere (PC-1) I want to say I already renewed so that VETC stops calling me this year. FR: FR-010, FR-036 · Priority: M

Scenario: Already renewed
 Given I am verified and my expiry on file is 2026-10-20
 When I say "tôi đã gia hạn rồi" and then "mua bên Bảo Việt, hết hạn tháng 9 năm sau"
 Then the outcome is "already_renewed"
 And my profile expiry becomes 2027-10-20 with method "voice_bot", confidence 0.6, insurer "OTHER" and a competitor note
 And my lead is recomputed

US-028: Stop calling me

As a customer (PC-1) I want to opt out by voice so that I never get sales calls again. FR: FR-036, FR-078 · Priority: M

Scenario: Opt-out
 When I say "đừng gọi nữa"
 Then the bot apologises and ends with outcome "opted_out"
 And my consent becomes call = false, dnc = true (persisting across reloads)
 And the audit trail has "consent.withdrawn" via the session id

US-029: Run an automated campaign

As a campaign manager (PS-03) I want to launch bot calls to the top hot leads so that humans only talk to interested customers. FR: FR-038 · Priority: M

Scenario: Eligible customers only

Given 50 hot leads of which 10 lack call consent, 2 are DNC and 5 are company-owned
 When I POST /api/voice/campaign with tier "hot" and limit 20
 Then at most 20 calls are made, none to the 17 ineligible customers
 And the response summarises outcomes (e.g. hot_handoff, link_sent, already_renewed)

Scenario: Contact policy applies to campaigns (target behaviour)

Given it is 21:00 ICT
 When I launch a campaign
 Then no calls are placed and the response explains "outside allowed contact hours"
 # Implemented: the campaign applies the full contact policy (doc 05 E-01, resolved)

US-030: Rehearse a call

As a telesales supervisor (PS-02) **I want** to play the customer in a console session **so that** I can QA script changes before activation. **FR:** FR-037 · **Priority:** S

Scenario: Console session

Given I hold "voice:operate"
 When I POST /api/voice/sessions for a non-DNC customer and send turns as text
 Then I see the Vietnamese bot lines with English gloss for each turn
 And when the call ends the outcome and any handoff id are shown

Scenario: DNC customer

When I start a session for a DNC customer
 Then the request is refused with "Customer is on the do-not-contact list"

US-031: Govern the bot

As a compliance officer (PS-05) **I want** bot governance KPIs **so that** I can show the regulator the bot is safe. **FR:** FR-039, FR-102 · **Priority:** S

Scenario: Governance dashboard

When I GET /api/dashboard/governance
 Then I see total calls, outcomes, plateVerificationFailureRate, optOutRate and the disclosure statement
 And the audit-chain verification result

EP-06 Telesales closing

US-032: Receive a warm handoff

As a telesales agent (PS-01) **I want** a handoff with everything I need and nothing more **so that** I can close in one call. **FR:** FR-040 · **Priority:** M

Scenario: Handoff content

Given a bot call ended with "hot_handoff" after the customer asked about price
 Then a handoff exists with plate, plateVerifiedByCustomer = true, masked phone, journey, expiry, premium and score
 And talking points include "Call from the official VETC hotline...", the regulated-price point and the customer's top benefits
 And the full phone number is visible only to users with profile:read_pii via Customer 360

US-033: Work my queue

As a telesales agent (PS-01) I want to claim, update and close handoffs so that the team doesn't call the same customer twice. FR: FR-041, FR-042, FR-093 · **Priority:** M

```
Scenario: Claim
  Given an open unassigned handoff
  When I PATCH it with status "claimed"
  Then it is assigned to me

Scenario: Invalid transition
  Given a handoff with status "won"
  When I PATCH status "open"
  Then the request fails with "Cannot move handoff from won to open"

Scenario: Someone else's handoff
  Given a handoff assigned to another agent
  When I GET it
  Then I receive 403 (policy agent_own_handoffs)
```

US-034: Balance the team

As a telesales supervisor (PS-02) I want to assign handoffs so that workload and skills are matched. FR: FR-043 · **Priority:** M

```
Scenario: Assign
  Given I hold "handoff:assign"
  When I PATCH a handoff with assignTo "U-agent01"
  Then it is assigned and the audit trail records "handoff.updated"

Scenario: Agent cannot assign
  Given I am a telesales agent
  When I PATCH with assignTo
  Then I receive 403 "Missing permission handoff:assign"
```

US-035: Close without taking payment by phone

As a telesales agent (PS-01) I want to build the right quote and send it to the customer's VETC app while on the call so that the customer pays safely in the app. FR: FR-045, FR-046, FR-047, FR-051, FR-077 · **Priority:** M

```
Scenario: Quote on the call, pay in app
  Given the customer agrees to renew with passenger cover
  When I POST /api/quotes with TNDS_CAR and PA_SEAT (5 seats, 20,000,000 per seat), channel "telesales"
  Then the quote shows 480,700 + 100,000 = 580,700 VND and bundle "SAFE_DRIVE"
  When I POST /api/quotes/{id}/send
  Then the quote is sent to the customer's VETC app or Zalo as a service message
  And the customer sees it in GET /api/customer/quotes
  And I mark the handoff "won" only after the customer pays in the VETC app

Scenario: Payment is customer-confirmed (target behaviour)
  Then there is no staff payment endpoint (POST /api/orders returns 404)
  And only the customer can pay, with POST /api/customer/orders and an Idempotency-Key
  # Implemented (doc 05 E-22, resolved)
```

EP-07 Customer app: renew and self-service

US-036: Open my renewal from a message

As a customer (PC-1) I want the link in my reminder to open my vehicle directly **so that** I don't have to log in or search. **FR:** FR-074, FR-095 · **Priority:** M

Scenario: Signed link
 Given a reminder link containing r=<profileId>.<signature>
 When the app POSTs it to /api/customer/session
 Then a 1-hour customer token scoped to my profile is returned

Scenario: Tampered link
 When the signature does not match
 Then the response is 401 "Link invalid or expired"

US-037: See my cover at a glance

As a customer (PC-1) I want a status card with my expiry, insurer and price **so that** I know whether I need to act. **FR:** FR-075 · **Priority:** M

Scenario: Low confidence asks for confirmation
 Given my expiry confidence is 0.5
 When I open the home screen
 Then the cover card shows the expiry with needsConfirmation = true and a "Confirm my expiry" action
 And up to 3 approved benefits with reasons are shown

US-038: Confirm my expiry

As a customer (PC-1) I want to confirm or correct my expiry date in the app **so that** VETC reminds me at the right time. **FR:** FR-076 · **Priority:** M

Scenario: Declare expiry
 When I POST /api/customer/expiry with expiryDate "2027-02-14" and insurer "PTI"
 Then my profile expiry becomes "2027-02-14" (customer_declared) and my journey and touchpoints are re-planned

US-039: Renew in one tap with my VETC wallet

As a customer (PC-1) I want to pay from my VETC wallet and get my certificate immediately **so that** renewing takes under a minute. **FR:** FR-046, FR-047, FR-049, FR-053, FR-056, FR-077 · **Priority:** M

Scenario: Happy path

Given my current TNDSCA ends 2026-11-30 and my category is car_under6
 When I request a TNDSCA_CAR quote
 Then the start date is 2026-12-01 and total is 480,700 VND (437,000 + 43,700 VAT)
 When I confirm with an Idempotency-Key
 Then my wallet is debited once, a policy with certificate number "TAS-..." is issued and my journeys stop

Scenario: Double tap

When the same order request is retried with the same Idempotency-Key
 Then the original order is returned with idempotentReplay = true and no second debit occurs
 And if the earlier attempt failed, reusing its Idempotency-Key returns 409

Scenario: Expired quote

Given the quote was created more than 24 hours ago
 Then purchase fails with "Quote expired – please re-quote"

US-040: Prove I am insured

As a customer (PC-1) I want a QR on my e-certificate that anyone can scan **so that** police or an inspection centre can check my cover. **FR:** FR-027, FR-050 · **Priority:** M

Scenario: Verification without PII

Given my certificate "TAS-TNDSCA-2026-100123" is active today
 When anyone opens /verify/TAS-TNDSCA-2026-100123
 Then they see valid = true, product, masked plate "30A-***.45", insurer "TASCO Insurance", start and end date
 And no name or phone is shown

Scenario: Unknown certificate

Then valid = false with reason "not_found"

US-041: Buy for several years

As a customer (PC-1) I want to buy 2–3 years aligned with my inspection cycle **so that** I don't have to renew every year. **FR:** FR-046, FR-053 · **Priority:** S

Scenario: Multi-year

When I request a TNDSCA_CAR quote with termYears 3
 Then the premium equals annual × (days in term ÷ 365) with VAT 10%, and the end date is 3 years after start

Scenario: Out of range

When I request termYears 4
 Then the request fails validation (1..3)

US-042: Don't lose my money if issuance fails

As a customer (PC-1) I want an automatic refund if the certificate cannot be issued **so that** I'm never charged for nothing. **FR:** FR-048 · **Priority:** M

Scenario: Issuance failure after payment

Given the wallet debit succeeded and TASCO core issuance fails
 Then any issued lines are cancelled, the payment is refunded in full, the order status is "issuance_failed_refunded" and "order.issuance_failed" is audited

EP-08 Value beyond discount and cross-sell

US-043: Relevant benefits, not discounts

As a long-haul driver (PC-2) **I want** to see the benefits that matter to my driving **so that** I see value in renewing via VETC. **FR:** FR-058, FR-059 · **Priority:** M

Scenario: Personalised ranking
 Given I drove 2,400 km on highways in 90 days and made 40 toll trips in 30 days
 Then "24/7 roadside assistance" ranks first with why "drives 2400 km on highways per quarter"

Scenario: Legal gating
 Given "loyalty_points" has legalStatus "pending_legal_review"
 Then it is never returned by /api/customer/home
 But staff see it flagged in Customer 360

US-044: Protect my passengers in the same checkout

As a family car owner (PC-1, 7 seats) **I want** to add passenger accident cover **so that** my family is protected, which TNDS does not do. **FR:** FR-046, FR-055 · **Priority:** S

Scenario: PA add-on
 When I quote TNDS_CAR with PA_SEAT (7 seats × 10,000,000)
 Then the PA line premium is 70,000 VND with VAT 0
 And the benefit "upsell_pa_seat" explains "family-size vehicle – passengers are not covered by TNDS"

US-045: Quote own-damage cover

As a new car buyer (PC-3) **I want** a physical-damage quote in two taps **so that** my new car is protected. **FR:** FR-054 · **Priority:** S

Scenario: PD rating
 Given my car is 2 years old, personal use
 When I quote MOTOR_PD with sumInsured 700,000,000 and deductible 1,000,000
 Then net premium is $700,000,000 \times 1.5\% \times (1 - 10\%) = 9,450,000$ VND plus 10% VAT
 And payment is blocked until a vehicle inspection is recorded with POST /api/quotes/{id}/inspection

Scenario: Referral above the online limit
 When sumInsured is 6,000,000,000
 Then the quote fails with "Sum insured exceeds the online limit – refer to underwriter"

US-046: Offer the upgrade after purchase

As a campaign manager (PS-03) **I want** a cross-sell message the day after a TNDS-only purchase **so that** we lift value per customer. **FR:** FR-028 · **Priority:** S

Scenario: Cross-sell scheduled
 Given a customer with marketing consent bought TNDS only
 When policy.issued is processed
 Then a "cross_sell" touchpoint is scheduled for tomorrow on app_push/zaio_zns

Scenario: Already bought add-ons or no consent
 Given the order included PA_SEAT, or consent.marketing = false
 Then no cross_sell touchpoint is scheduled

US-047: Loyalty and referral only when lawful

As a compliance officer (PS-05) **I want** loyalty and referral switched off until legal approves **so that** no unlawful inducement reaches customers. **FR:** FR-060 · **Priority:** C

```
Scenario: Referral disabled
  Given referral.enabled = false and legalStatus "pending_legal_review"
  Then no referral offer appears in any customer channel
  When legal approves and a new referral version with enabled = true is approved by a second user
  Then referral rewards are non-cash VETC points only, max 5 referrals per month
```

US-048: Never lapse with auto-renew (roadmap)

As a customer with wallet auto top-up (PC-2) **I want** to opt in to auto-renewal **so that** I am never uninsured. **FR:** FR-061 · **Priority:** S

```
Scenario: Confirm-before-debit
  Given I opted in to auto-renew
  When my expiry is 7 days away
  Then I receive a confirmation request in the VETC app
  And my wallet is debited only after I confirm
  And if I do not confirm, no debit happens and normal reminders continue
  # Auto-renew is on the roadmap (benefit available = false). The automated test verifies the
  # confirm-before-debit guarantee today: staff cannot debit a customer's wallet.
```

EP-09 Partner channel

US-049: Onboard a partner

As a partner manager (PS-08) **I want** to onboard a bank or showroom and give them an API key **so that** they can sell TASCO cover from their own systems. **FR:** FR-063, FR-064, FR-095 · **Priority:** M

```
Scenario: Onboard and issue key
  When I POST /api/partners with name "ABC Showroom", type "showroom", region "Hà Nội"
  Then a partner "P-XXXXXXX" with status "active" exists
  When I POST /api/partners/{id}/keys
  Then a key starting "tpk_" is shown once, only its hash and an 8-character prefix are stored, and
  "partner.api_key_issued" is audited
```

US-050: Quote a brand - new car at the showroom

As a showroom salesperson using the partner API (PS-09) **I want** to quote TNDS by plate at delivery **so that** the customer drives away insured. **FR:** FR-065 · **Priority:** M

```
Scenario: Unknown plate becomes a new profile
  Given plate "51K-888.99" is not in the VETC base
  When I POST /api/partner/v1/quotes with plate, holderName, phone, seats 5, usage "personal",
  consentMarketing true and product TNDS_CAR
  Then a partner_showroom source record is ingested and a profile "51K88899" is created
  And a quote with channel "partner_api" and my partnerId is returned
```

```
Scenario: Product not sold on partner channel
  When I quote TNDS_MOTORBIKE
  Then the request fails with "TNDS_MOTORBIKE is not sold on channel partner_api"
```

US-051: Bind safely

As a bank partner integration (PS-09) I want idempotent binding **so that** network retries never create duplicate policies. **FR:** FR-047, FR-066 · **Priority:** M

```
Scenario: Own quote
  When I POST /api/partner/v1/orders with my quoteId and an Idempotency-Key
  Then the policy is issued and the order records my commission lines
  And no VETC wallet is debited: the payment reference starts with "PARTNER-" (premium collected by the partner)

Scenario: Another partner's quote
  When I bind a quote created by another partner
  Then I receive 404 "Quote not found"
```

US-052: Transparent commission

As a partner (PS-09) I want my commission statement for any period **so that** reconciliation with TASCO is effortless. **FR:** FR-067, FR-068, FR-069 · **Priority:** M

```
Scenario: Statement
  Given I sold one TND5_CAR (net 437,000) and one PA_SEAT (net 100,000) as a showroom
  When I GET /api/partner/v1/statement?from=2026-10-01&to=2026-10-31
  Then lines show TND5 5% = 21,850 and PA 15% = 15,000, total 36,850 VND

Scenario: Cap enforced
  Given a draft commission rule sets TND5 rate 0.07
  Then validation fails with "exceeds statutory cap 0.05 for TND5_CAR"
```

US-053: Suspend a partner

As a partner manager (PS-08) I want to suspend a partner instantly **so that** misuse stops immediately. **FR:** FR-063 · **Priority:** M

```
Scenario: Suspension blocks API access
  When I PATCH /api/partners/{id} with status "suspended"
  Then subsequent calls with that partner's keys are rejected as unauthenticated
```

EP-10 Fleet / B2B

US-054: Route company vehicles to account management

As a fleet manager (PC-5) I want to deal with one account manager, not robocalls per vehicle **so that** renewal is managed for the whole fleet. **FR:** FR-015, FR-038, FR-070 · **Priority:** S

```
Scenario: Fleet routing
  Given a profile with ownerType "company"
  Then its NBA is "route_b2b"
  And it is skipped by /api/voice/campaign

Scenario: No B2C journeys for fleet (target behaviour)
  Then company-owned vehicles are not enrolled in any consumer journey and have no scheduled touchpoints
  # Implemented: journey audiences exclude ownerType = company (doc 05 E-07, resolved)
```

US-055: Fleet renewal dashboard (roadmap)

As a fleet manager (PC-5) I want one view of every vehicle's cover and one VAT invoice **so that** my admin cost drops. **FR:** FR-070 · **Priority:** S

```
Scenario: Fleet view
  Given my company has 40 vehicles in the VETC base
  When I open the fleet dashboard
  Then I see each plate's cover status and expiry, sorted by soonest expiry
  And I can renew a selection in one order and receive one consolidated VAT invoice
  # Roadmap (benefit fleet_dashboard available = false); recorded as a manual/roadmap test
```

EP-11 Claims FNOL

US-056: Report an accident from the app

As a customer (PC-1) I want to report an accident with photos and location **so that** TASCO helps me quickly. **FR:** FR-071 · **Priority:** S

```
Scenario: Valid FNOL
  Given I hold active policy "TAS-TNDSCA-2026-100123" valid from 2026-01-01 to 2026-12-31
  When I POST /api/customer/claims with incidentDate "2026-10-05", description, location and photos 3
  Then a claim "CL-XXXXXXXX" with status "submitted" and slaDueAt = now + 4 hours is created
  And an acknowledgement message ("claim_received") is sent to me

Scenario: Outside policy period
  When incidentDate is "2027-02-01"
  Then the request fails with "Incident date is outside the policy period"

Scenario: Not my policy
  When I reference another customer's policy
  Then I receive 404 "Policy not found"
```

US-057: Progress a claim

As a claims handler (PS-07) I want to move claims through clear statuses **so that** customers and auditors see progress. **FR:** FR-072 · **Priority:** S

```
Scenario: Valid transition
  Given claim status "submitted"
  When I PATCH status "acknowledged" with a note
  Then history records the transition with my id and "claim.status_changed" is audited

Scenario: Skipping steps
  When I PATCH a "submitted" claim to "paid"
  Then it fails with "Cannot move claim from submitted to paid"
```

US-058: Track my claim

As a customer (PC-1) I want to see my claim status **so that** I don't have to call the hotline. **FR:** FR-073 · **Priority:** S

```
Scenario: My claims only
When I GET /api/customer/claims
Then only claims for my profile are returned with current status and history
```

EP-12 Privacy and consent

US-059: Control how VETC contacts me

As a customer (PC-1) I want to switch marketing and call consent on or off **so that** I decide how I am contacted. **FR:** FR-078 · **Priority:** M

```
Scenario: Withdraw call consent
When I PUT /api/customer/consent with call = false
Then no voice_bot or telesales contact is made to me
And the change is audited as "consent.updated" and survives source reloads
```

US-060: Download my data

As a customer (PC-1) I want to download what VETC/TASCO hold about me **so that** I can exercise my right of access. **FR:** FR-079 · **Priority:** M

```
Scenario: Self-service export
When I GET /api/customer/data-export
Then I receive my profile, lead, policies, quotes, orders, claims, telesales tasks, messages,
source records (without internal hidden fields) and voice sessions
And "dsar.access_exported" is audited
```

US-061: Erase personal data lawfully

As a compliance officer (PS-05) I want to anonymise a data subject's personal data **so that** we honour erasure requests without breaking legal retention. **FR:** FR-080 · **Priority:** M

```
Scenario: Active policy blocks erasure
Given the customer has an active policy
When I POST /api/dsar/{id}/erase
Then it fails with "Active policy in force – personal data must be retained until expiry (legal obligation)"

Scenario: Erasure
Given no active policy
When I erase
Then name and phone are removed, consent becomes dnc, the lead is deleted, message text is "[erased]", transcripts are emptied
And PII in handoffs, claim text and voice signals is also scrubbed, and my sessions are invalidated
```

US-062: See only what my role needs

As a campaign manager (PS-03) without PII permission I want masked names and phones in Customer 360 **so that** I can do my job without exposing personal data. **FR:** FR-081, FR-097 · **Priority:** M

```

Scenario: Masking
  Given I lack "profile:read_pii"
  When I GET /api/customers/{id}
  Then name and phone are masked and piiMasked = true
  And "profile.viewed" is audited with piiVisible = false

```

EP-13 Rule governance

US-063: Change scoring without a release

As a rule author (PS-04) I want to edit scoring weights as a draft so that the business can tune prioritisation itself. FR: FR-057, FR-082, FR-083, FR-084 · Priority: M

```

Scenario: Invalid weights rejected
  When I POST /api/rules/validate for kind "scoring" with weights summing to 95
  Then errors include "factor weights must sum to 100 (got 95)"

Scenario: Valid draft
  When I POST /api/rules with a valid scoring payload
  Then a version "scoring@2" with status "draft" and a checksum is created and audited

```

US-064: Simulate before I submit

As a rule author (PS-04) I want to see how a change affects a real customer so that I avoid surprises. FR: FR-085 · Priority: S

```

Scenario: Current vs candidate
  When I POST /api/rules/simulate with profileId, kind "nba" and a candidate payload
  Then the response contains "current" and "candidate" evaluations (journey, score, tier, reasons, nextBestAction, benefits)

```

US-065: Four-eyes approval

As a compliance approver (PS-05) I want to approve or reject rule changes made by others so that no single person can change customer-facing behaviour. FR: FR-086 · Priority: M

```

Scenario: Self-approval forbidden
  Given "author" submitted "scoring@2"
  When "author" tries to approve it
  Then the response is 403 "Maker-checker: you cannot approve your own change"

Scenario: Approval activates and retires
  When "approver" approves with comment "OK per campaign review"
  Then "scoring@2" is active, "scoring@1" is retired with supersededBy "scoring@2"
  And all replicas use the new version within 15 seconds

```

US-066: Roll back quickly

As a rule author (PS-04) I want to roll back to an earlier version so that a bad change can be reversed safely. FR: FR-087 · Priority: M

Scenario: Rollback is a governed change
 When I POST /api/rules/scoring@1/rollback
 Then a new draft "scoring@3" with the payload of scoring@1 is created
 And it requires submission and approval by another user

US-067: Copy guard on content changes

As a compliance approver (PS-05) **I want** content drafts with banned phrases to be impossible to save **so that** I never need to catch them by eye. **FR:** FR-024, FR-083 · **Priority:** M

Scenario: Banned phrase in template
 When I create a "content.messages" draft containing "Giảm giá 10% khi gia hạn"
 Then the draft is rejected with a copy guard error naming "giảm giá"

US-068: Commission within the law

As a partner manager (PS-08) **I want** commission rules validated against statutory caps **so that** we never overpay. **FR:** FR-068, FR-083 · **Priority:** M

Scenario: Cap
 When a commission draft sets PA_SEAT rate 0.25 with cap 0.2
 Then validation fails

EP-14 Identity, access and audit

US-069: Strong sign-in for privileged roles

As a rule approver (PS-05) **I want** MFA on my account **so that** approvals cannot be forged with a stolen password. **FR:** FR-089, FR-090, FR-091 · **Priority:** M

Scenario: Two-step login
 When I POST /api/auth/login with correct credentials
 Then I receive mfaRequired = true and a 5-minute mfaToken
 When I POST /api/auth/mfa with a valid TOTP code
 Then I receive a 30-minute access token with amr ["pwd","otp"]

Scenario: Privileged role without MFA enrolment
 Given a new data_steward account that has not enrolled MFA
 When I POST /api/auth/login with correct credentials
 Then I receive mfaEnrolment = true and an otpauth URI shown only to me
 And my first valid TOTP code completes enrolment (codes cannot be reused)

US-070: Lock out brute force

As a security officer (PS-12) **I want** accounts locked after repeated failures **so that** password guessing fails. **FR:** FR-089 · **Priority:** M

Scenario: Lockout
 When 5 wrong passwords are submitted for "agent.hn"
 Then the 6th attempt within 15 minutes returns "locked" (HTTP 423) even with the right password
 And wrong MFA codes count towards the same lockout
 And each failure is audited as "auth.login_failed"

US-071: Manage users with segregation of duties

As a platform admin (PS-12) I want to create users with roles and regions so that access matches the job. FR: FR-092, FR-094, FR-096 · **Priority:** M

```
Scenario: Create agent
  When I POST /api/users with username "agent.dn", roles ["telesales_agent"], region "Đà Nẵng" and a
  14-character password
  Then the user exists and "user.created" is audited
  And the user must change the password at first sign-in

Scenario: Weak password
  When the password has fewer than 12 characters
  Then validation fails with "at least 12 characters"

Scenario: Admin cannot read PII or approve rules
  Given I am admin
  When I GET /api/customers/{id} or approve a rule
  Then I receive 403
```

US-072: Prove the audit trail is intact

As an internal auditor (PS-13) I want to verify the audit hash chain so that I can attest nothing was altered. FR: FR-098, FR-099 · **Priority:** M

```
Scenario: Verify
  When I GET /api/audit/verify
  Then the result is ok = true with the number of entries and the head hash

Scenario: Tampering detected
  Given an audit entry's details were altered in the database
  Then verify returns ok = false with brokenAt = <index> and reason "hash mismatch"
```

EP-15 Operations, insights and UX

US-073: See growth and economics at a glance

As an executive (PS-10) I want a one-page dashboard of policies, premium, channels and cost so that I can steer investment. FR: FR-100 · **Priority:** M

```
Scenario: Overview
  When I GET /api/dashboard/overview
  Then I see base size, expiring in 30 days, lapsed (<= 60 days), orders and premium by journey and
  channel, active policies by product
  And economics: voice bot cost = calls × 1,500 × 1.6, telesales-equivalent = calls × 6,000 × 4.5,
  and the saving
```

US-074: Keep integrations healthy

As a support engineer (PS-11) I want integration status and reconciliation on demand so that I can find and fix issues before customers notice. FR: FR-052, FR-103, FR-104 · **Priority:** M

Scenario: Status

When I GET /api/ops/status
 Then I see circuit state for vetc-wallet, tasco-core, voice-ai, app-push, zalo-zns and sms, active rule versions with checksums and the event backlog

Scenario: Reconciliation

When I POST /api/ops/jobs/reconciliation
 Then a job run is recorded with checked count and any mismatches (missing payment ref, missing policy, stuck > 1h)

US-075: Apply data retention

As a compliance officer (PS-05) I want retention applied automatically **so that** we don't keep personal data longer than needed. **FR:** FR-105 · **Priority:** M

Scenario: Retention job

When the retention job runs
 Then source records older than 365 days and voice sessions older than 180 days are deleted
 And the job result lists each entity's action and cutoff date

US-076: Work in my language

As a telesales agent (PS-01) I want to switch the console between Vietnamese and English **so that** I work comfortably and can share screens with English-speaking reviewers. **FR:** FR-108, FR-110 · **Priority:** M

Scenario: Language toggle

When I switch to "VI"
 Then all labels, help text and dates (dd/mm/yyyy) appear in Vietnamese without reloading
 And my choice is remembered on this device

US-077: Verify a certificate at the roadside

As a traffic police officer or inspection-centre clerk (external) I want to scan the QR and see validity instantly **so that** I can trust a digital certificate. **FR:** FR-050, FR-109 · **Priority:** M

Scenario: Scan

When I scan the QR on the e-certificate
 Then /verify/{certNo} shows VALID or NOT VALID, product, masked plate, insurer and validity dates, in Vietnamese by default
 And the page works on a basic mobile browser without login